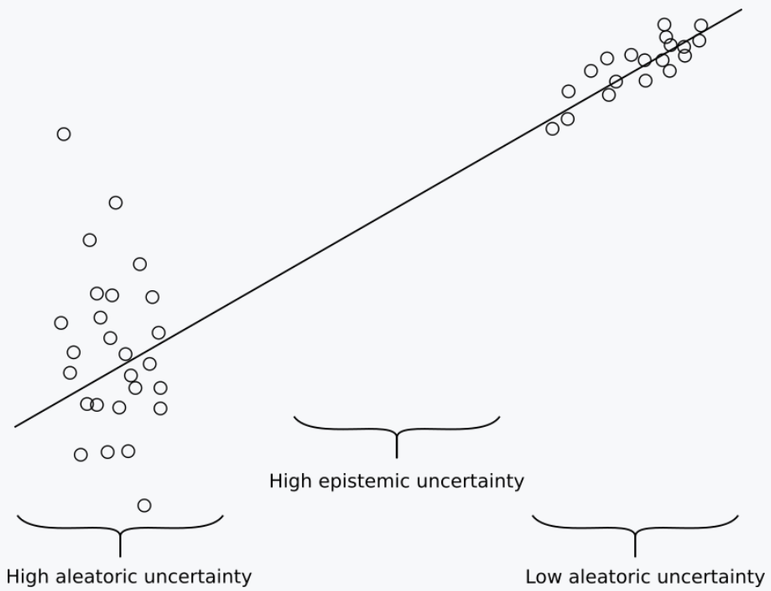
Introduction – what can be forecasted

WHAT CAN BE FORECASTED EASILY? 1/1

1. daily electricity demand in 3 days' time
2. Google stock price tomorrow
3. Google stock price in 6 months' time
4. maximum temperature tomorrow
5. total sales of drugs in Australian pharmacies next month

SOMETHING IS EASY TO FORECAST IF: 1/1

1. we have a good understanding of the factors that contribute to it
2. there is a lot of data available
3. the future is somewhat similar to the past
4. ID assumption: samples are identically distributed
5. the forecasts must not affect the quantity we are trying to forecast.
6. self-fulfilling prophecies (election polls)
7. controlled systems



Aleatoric (alea=latin for dice) or data uncertainty. This is the **irreducible** uncertainty that we cannot reduce by increasing observations.

Epistemic (epistḗmē=science) or model uncertainty. This uncertainty can be reduced by increasing observed data, and as such, can be **reduced**

Example

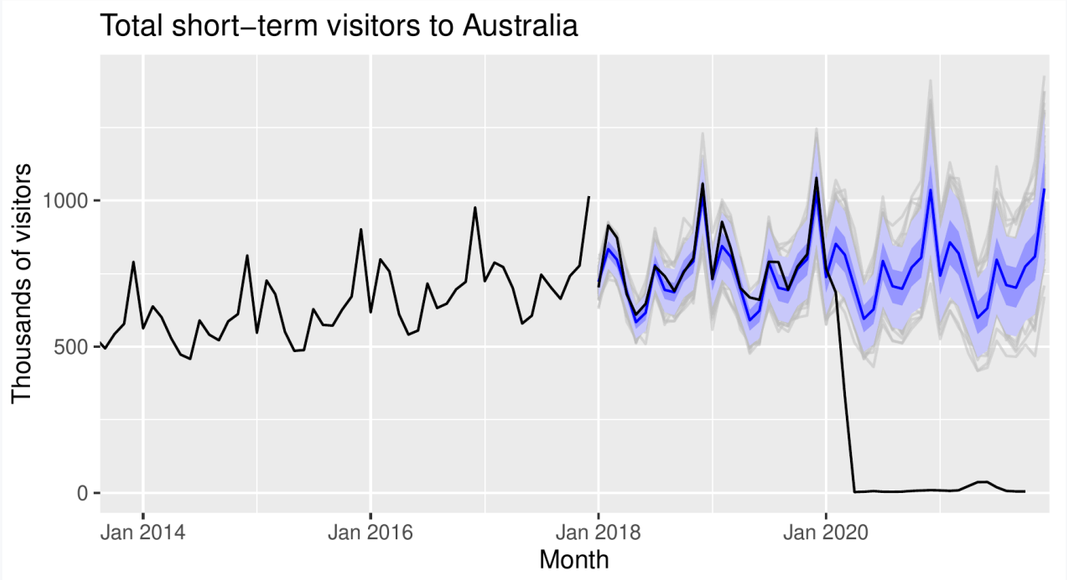
We can always decompose a signal in two parts: a **modeled** signal and an **error**

$$y = \hat{y} + \epsilon = f(x, \theta) + \epsilon$$

The set of parameters θ must be identified. Increasing the amount of data, our estimation can improve, that is, the associated error is epistemic. The modeling error ϵ can be due to:

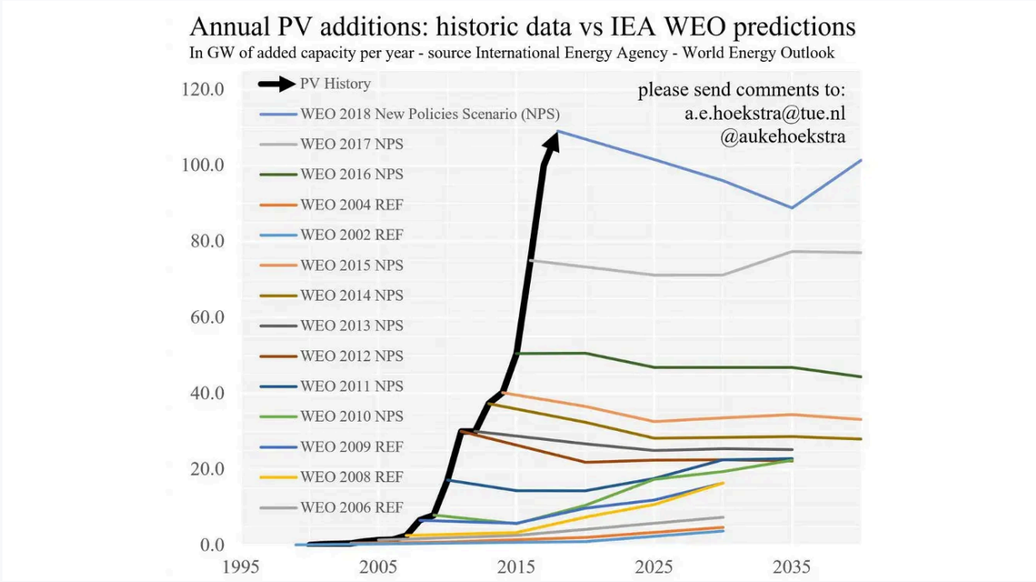
- disregarded inputs in the matrix of signals x
- wrong choice of the model f or its over simplicity
- irreducible errors associated to measurement noise

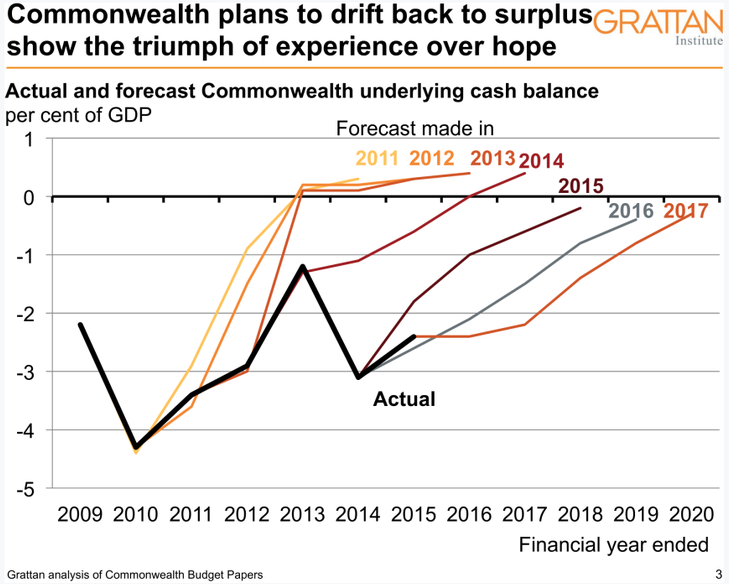
Aleatoric or epistemic uncertainty?



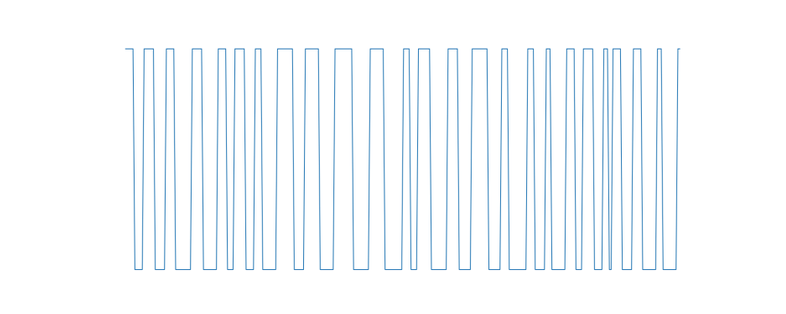
MODEL CHOICE 1/1

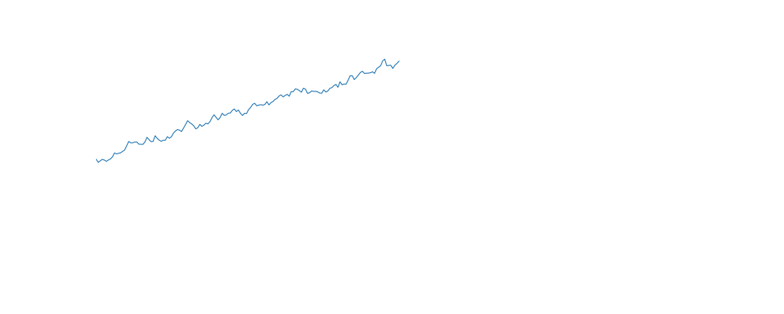
Model choice is another source of uncertainty, that can be mitigated by model selection. Unfortunately, it cannot correct ideological biases.

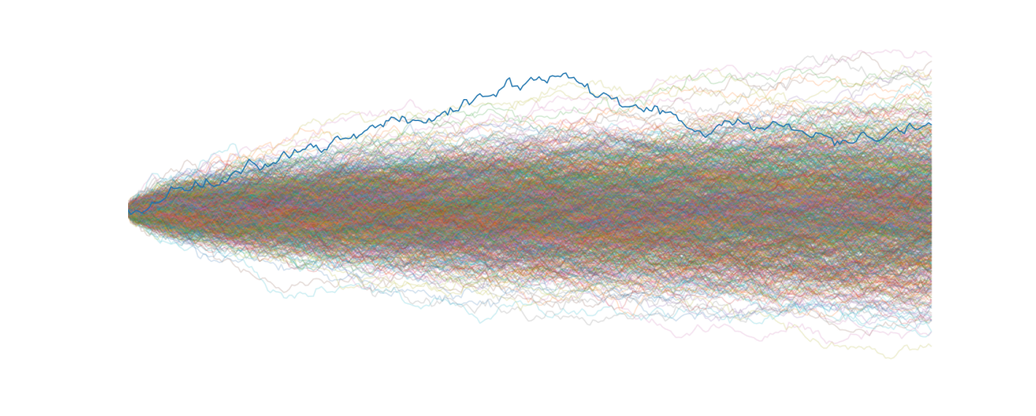




If some conditions are met, we can make informative decisions even if the underlying processes are not forecastable. Imagine you are a trader, which of the following signals you would like to forecast?





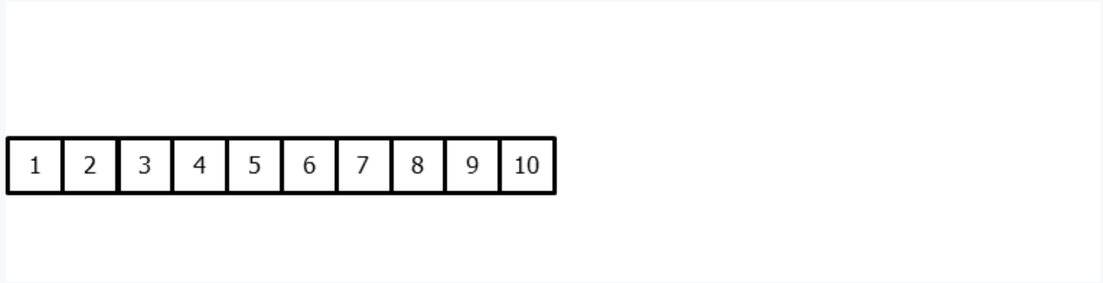


In this course we think to a time series as a function of time. We we'll use equivalently the expression $x(t)$ and x_t to represent a signal at time t. While the first expression is more typical of stochastic process literature, the second is more natural when referring to signals as discrete vectors.

$$x(t) = x_t \qquad x_t \in \mathbb{R}^d$$

Embeddings

With the word embedding we refer to the practice of rearranging a time series x into a matrix, or a tensor, by stacking windows of the signal. This is a very common practice in time series analysis since usually values of the signals close in time are “similar”.



Single vs multi-step forecasts

Most of the forecasting methods exploit information contained in the most recent historical values to predict the future. We usually use the notation $x_{t|t-k}$ to indicate the k most recent values

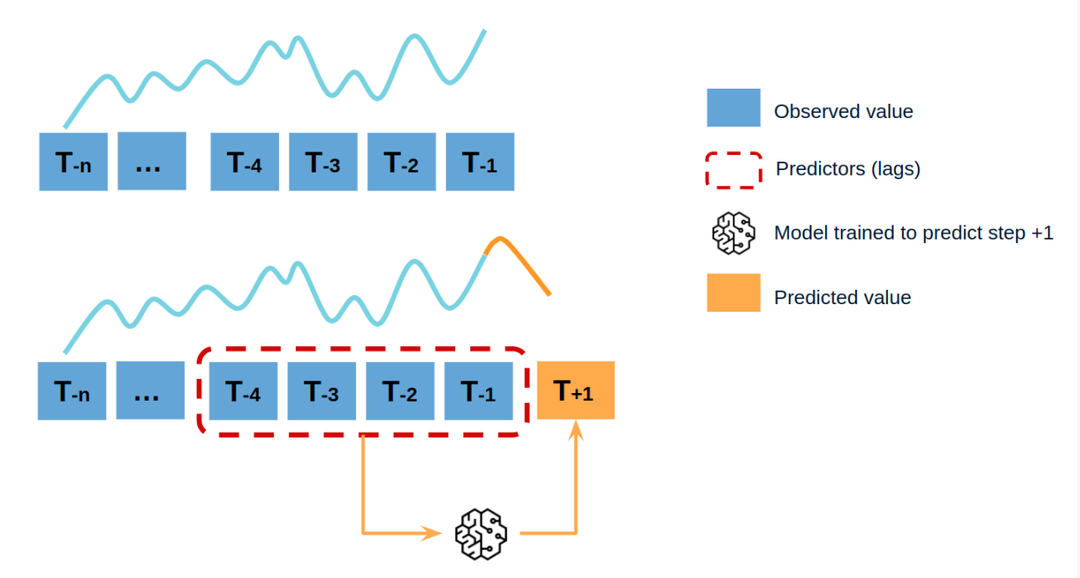
$$x_{t|t-k} = [x_{t-k}^T, x_{t-k+1}^T, ...x_t^T]^T$$

A forecasting function predicting the next step of the sequence can then be written:

$$\hat{x}_{t+1} = f(x_{t|t-k})$$

where $\hat{x}$ stands for a forecasted value of the time series x.

We're usually interested in predicting multiple steps ahead of the time series. Two main strategies are transversal to most of forecasting methodologies:



Recursive predictions

In recursive prediction the forecasting model is trained to predict just one step ahead $\hat{x}_{t+1}$, and recursively substitute the prediction of the next step in its feature set.

This is the strategy adopted by

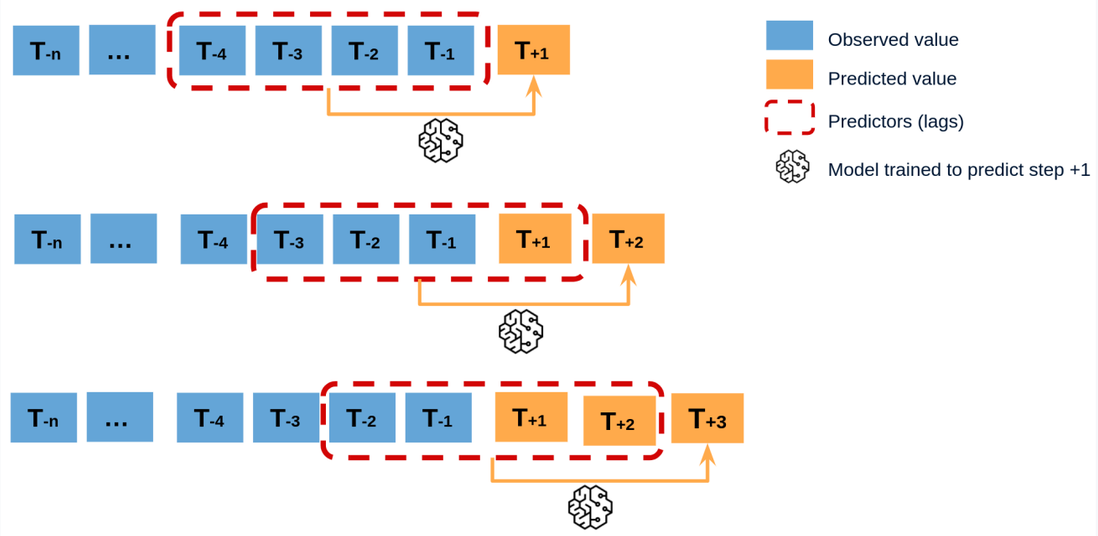
- exponential smoothing
- ARIMAX
- state space models

Machine learning methods can also use this strategy

✅ fit just one model

✅ simple implementation

❗ prediction errors tend to accumulate, stability is usually enforced



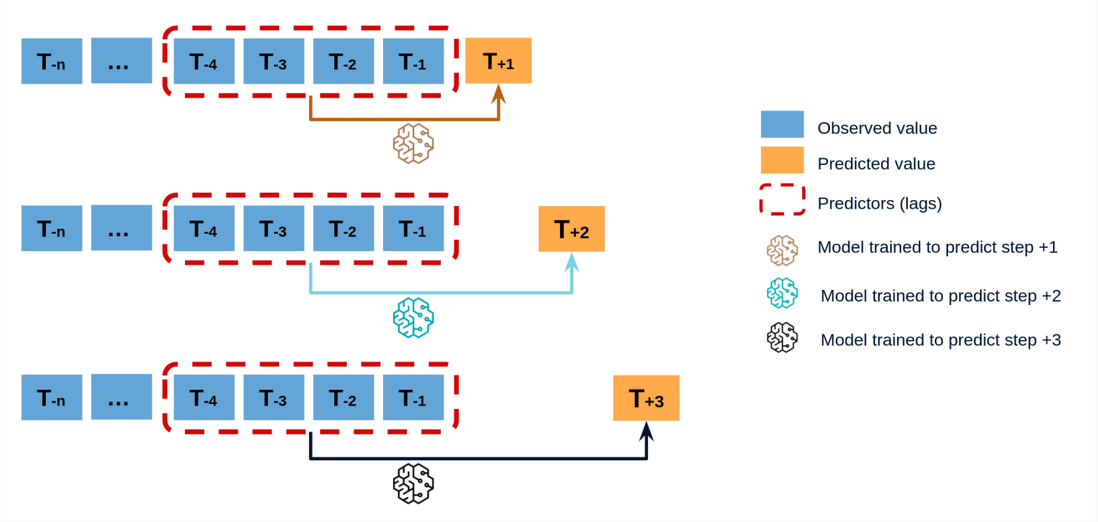
Direct multistep prediction

We fit multiple models, each one for a different step ahead

$$\hat{x}t_{+1} = f_1(x_{t|t-k})$$
$$\hat{x}t_{+2} = f_2(x_{t|t-k})$$
$$\dots$$
$$\hat{x}t_{+T} = f_T(x_{t|t-k})$$

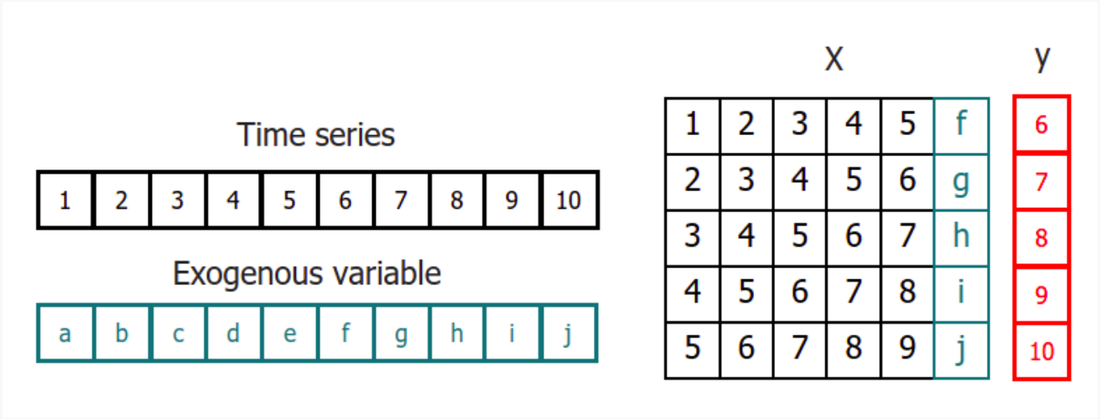
✅ simple implementation

❗ temporal consistency could be lost since models are independen



Endogenous and exogenous features

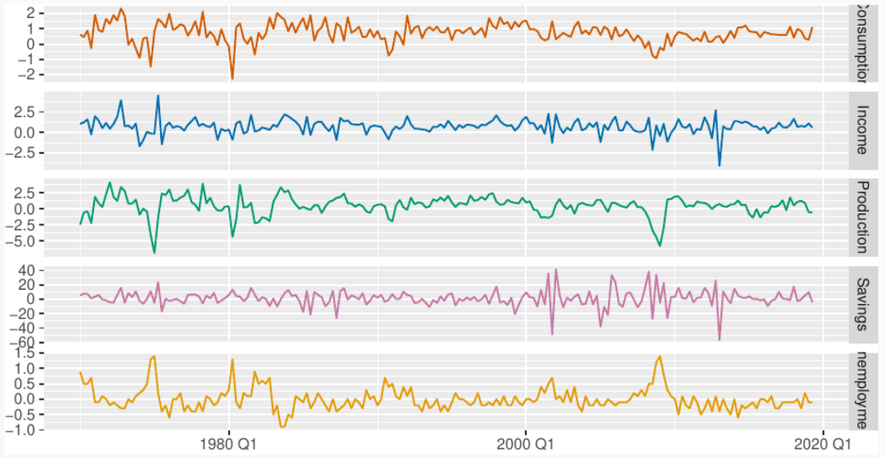
- endogenous: features crafted from the same time series we're trying to predict, e.g. lags or moving averages of the signal
- exogenous: other signals, metadata, categorical features that might be relevant for the prediction

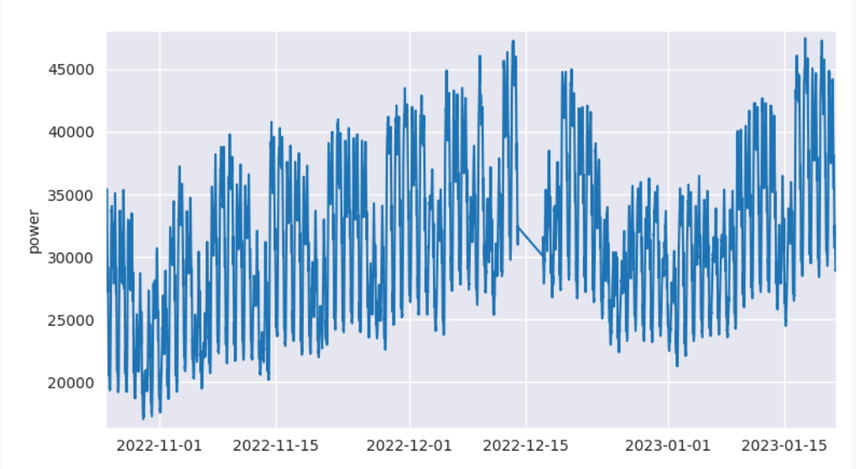


Line plots

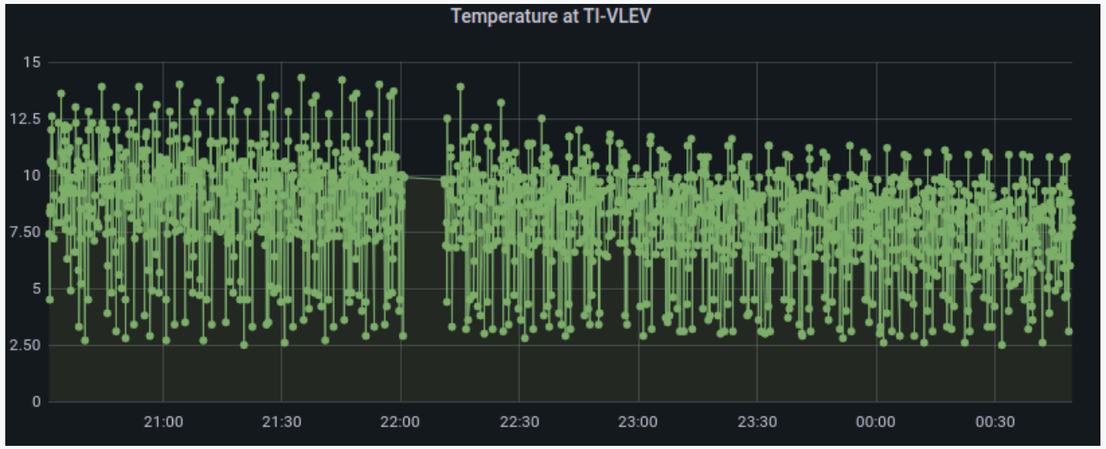
Line plots are the first visual aid in time series data analysis. T

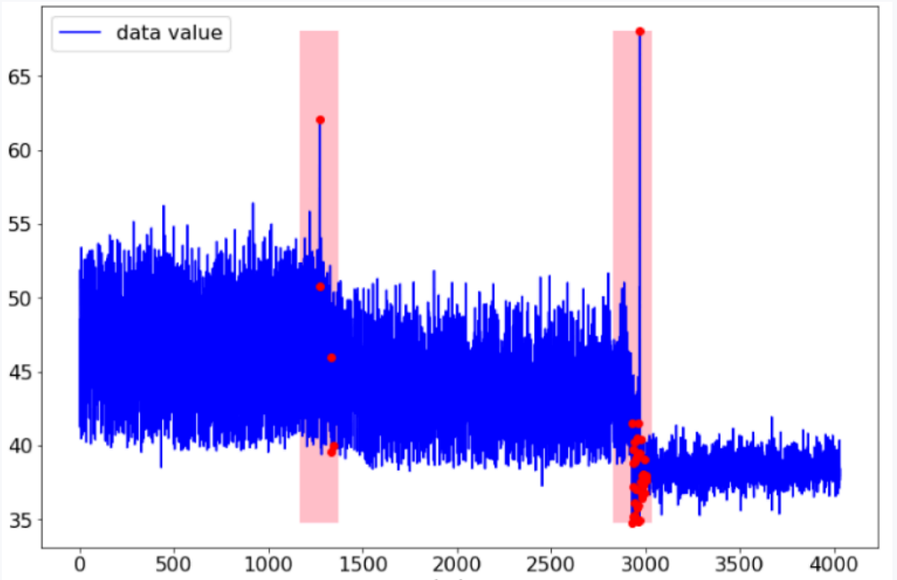
- check time series characteristics and regularity
- outliers
- interpolation errors
- …holes!





Line plots - continued



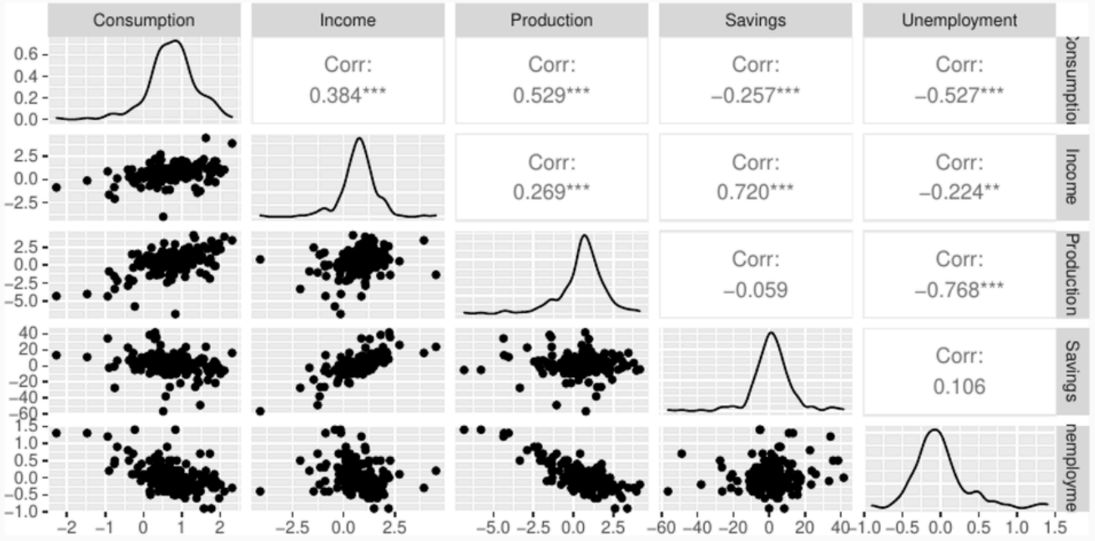


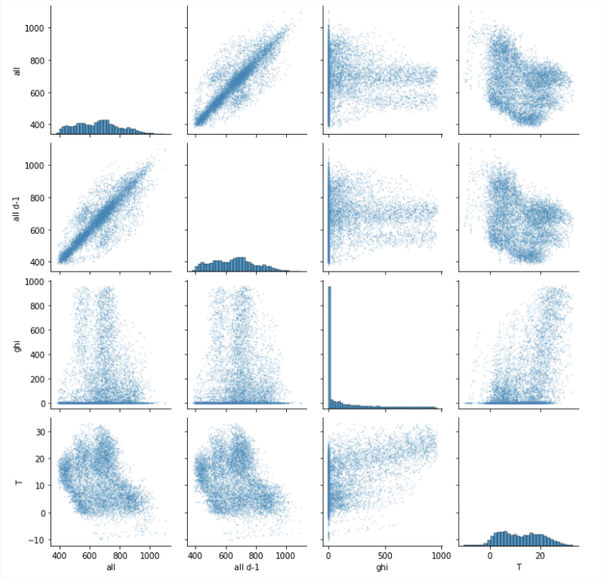
We often have datasets with too many time series to be plotted together

In this case, we can rely on quantile plots or box-plots

Scatter plots

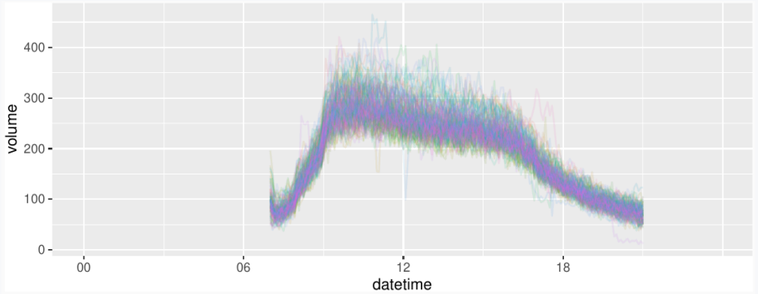
A scatter plot can be used to visually assess the dependence of the target y_t from exogenous inputs/features $x_{i,t}$.

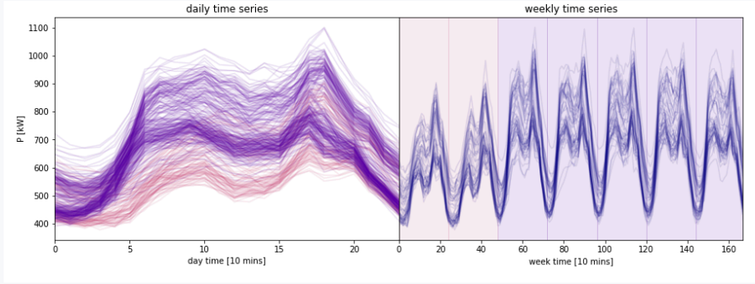


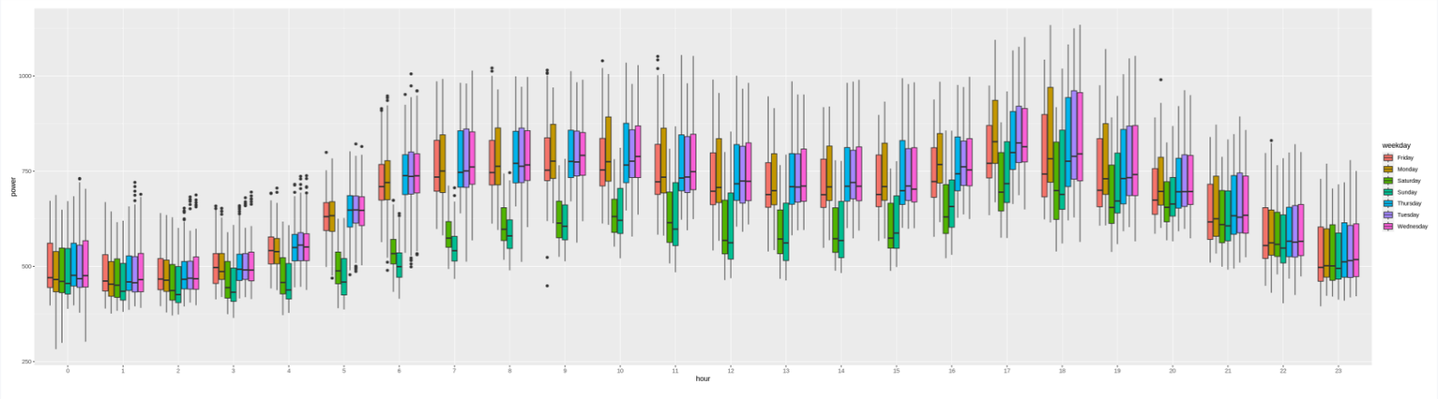


Embeddings

Time embedding is helpful to see whether one can reduce the variance of segments in the time series to be predicted.

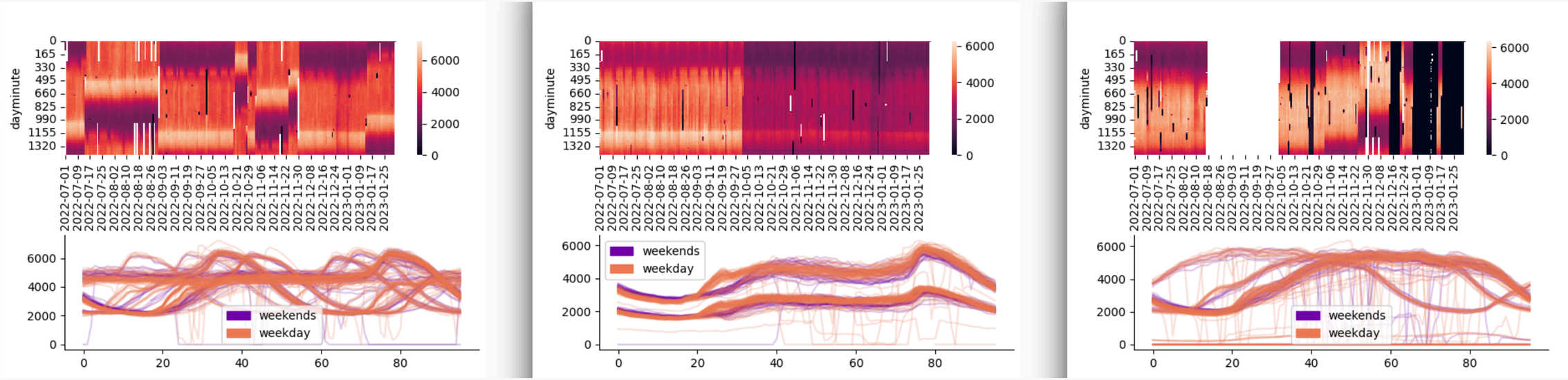






Embeddings - continued

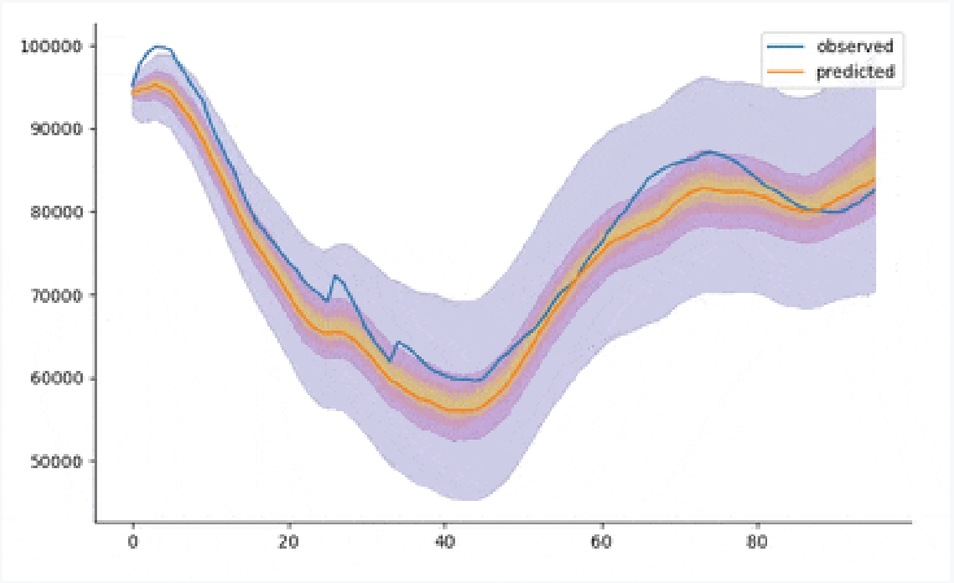
What is wrong with these time series of power measurements?



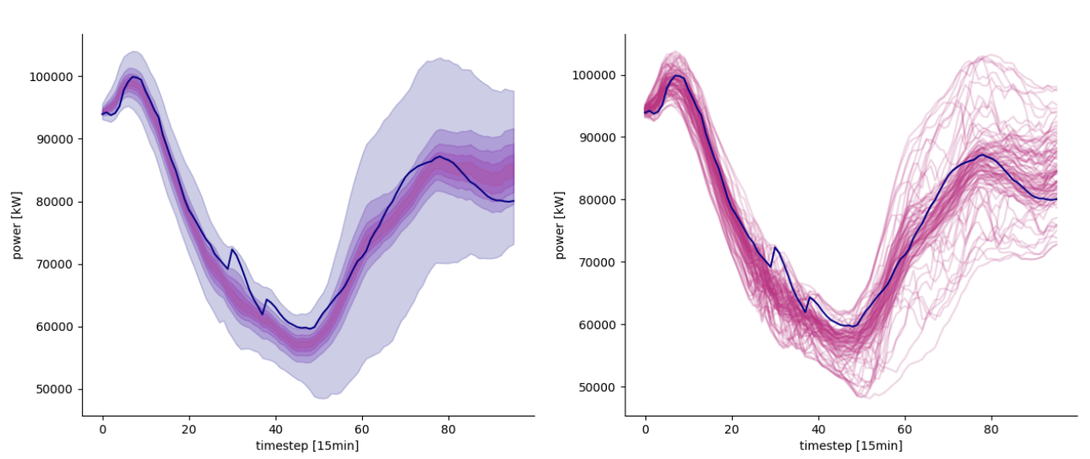
Power measurements from MV stations in Kenya. Top: heatmap embedded as daytime-day. Bottom: embedded time series plots.

Animations

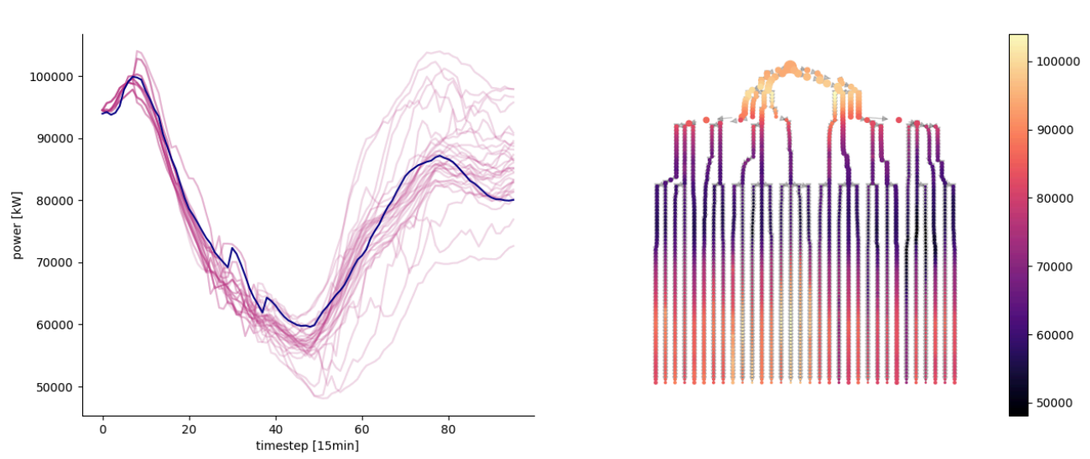

Besançon
Zurich
Innsbruck
Liechtenstein
Bern
Switzerland
Bolzano
Geneva
Sondrio
Trento
Annecy
Como
Aosta
Bergamo
Biella
Brescia
Verona
17:25:12pm - 18:34:12pm
Index
Y Axis
12 PM
06 PM
Jun 13
06 AM
12 PM
06 PM
Mon 14
06 AM


observed
predicted
100000
90000
80000
70000
60000
50000
0
20
40
60
80

Probabilistic predictions



Left: predicted quantiles. Right: 100 different scenarios



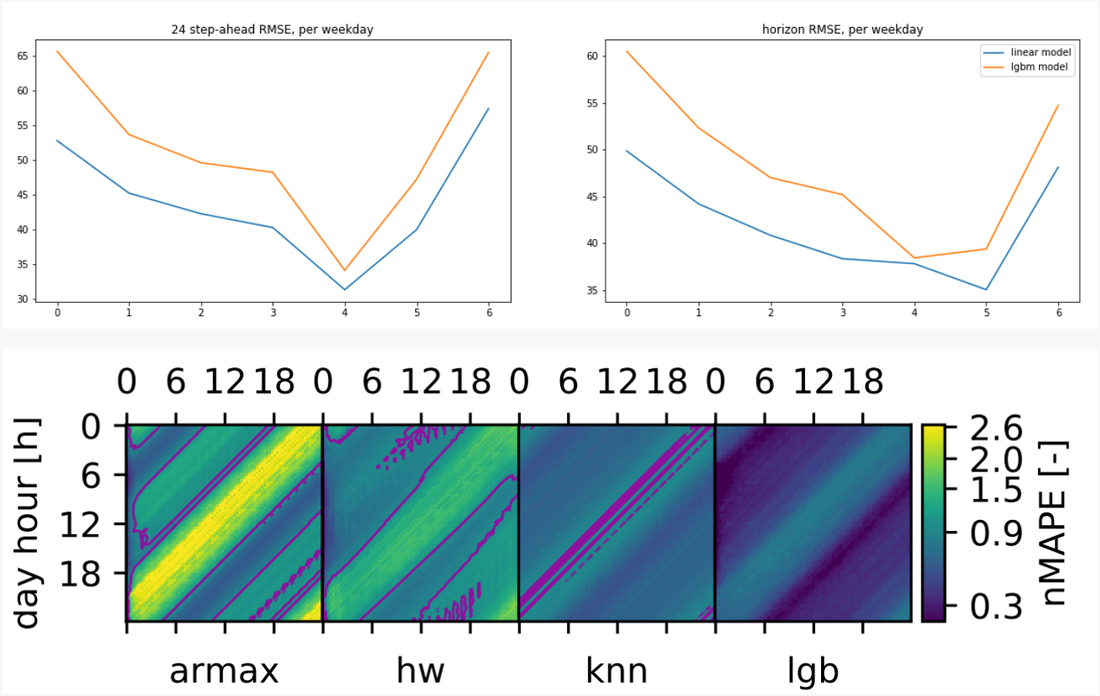
Left: a stochastic scenario tree obtained from the scenarios. Right: graph representation of the tree

Model diagnostics

Since we are usually interested in more than one step ahead forecasts, we need good ways to visualize our model performances in more than one dimension.

Given a loss $l(y_{t+k}, \hat{y}_{t+k})$, where k is the step ahead, we can decide to investigate:

- the loss as a function of k
- the average loss over $k \in [1, ...T]$ prediction steps
- the previous two points, but as a function of the time in which the prediction was made



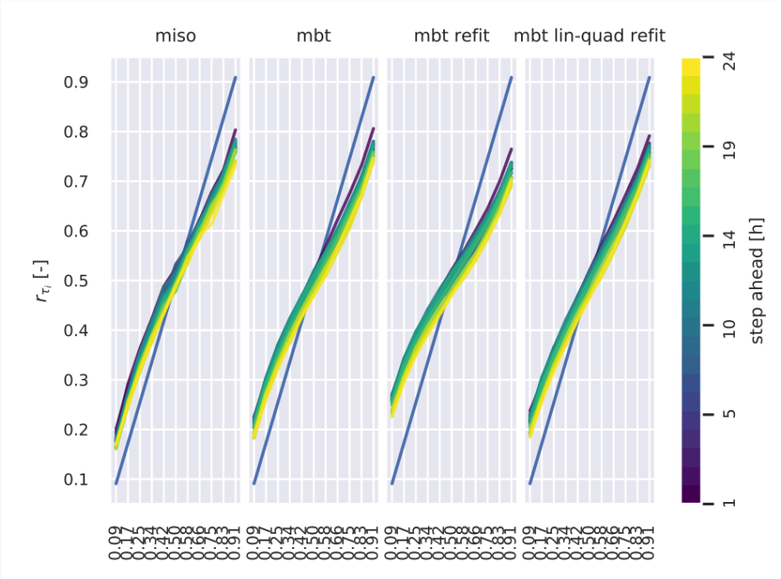
A loss function binned for time of the prediction and forecast's hour ahead.

Probabilistic scores

When producing probabilistic forecast is usually a good idea to plot an additional dimension that refers to the level of the quantile we're predicting.



Quantile loss functions for different quantile levels and different step ahead.



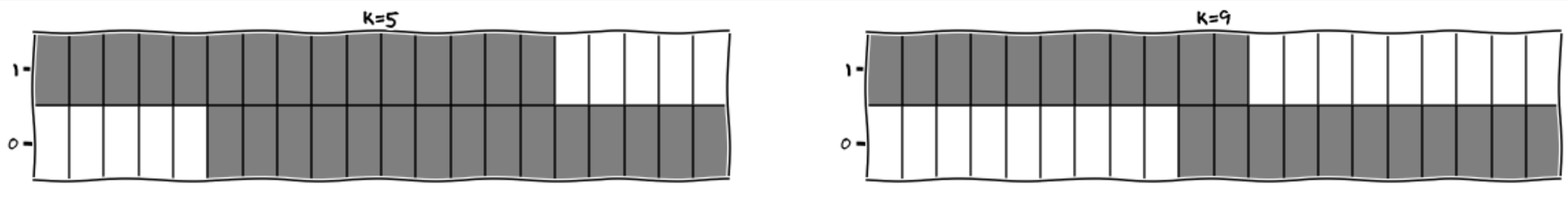
Reliability plot for different quantile levels and different step ahead

ACF function

The autocorrelation function is defined as:

$$\rho_k = \frac{\gamma_k}{\gamma_0} = \frac{cov(x_t, x_{t-k})}{\sigma(x_t)\sigma(x_{t-k})} \overset{stationary}{=} \frac{cov(x_t, x_{t-k})}{\sigma^2(x)}$$

If we get only T timesteps to evaluate ρ_k, we must be careful to keep $T - k$ large enough for the estimation to be meaningful



When evaluating ACF for lag 9 we just got 2 observations to estimate the correlation for a 20-steps time series

Which is which?

